■■ RAKSHAPATH (■■■■■■■) — TECHNICAL DOSSIER

STAGE 2: TECHNICAL PITCH, WORKING PROTOTYPE & 5-MINUTE VIDEO SUBMISSION GUIDE

Hackathon Stage: Screening Round 2	**Team Name:** Techies	**Target Track:** Diploma & Higher Education
Submission Format: 5-Min Video + PDF Evidence	**Deployment Status:** Vercel Edge Live + FastAPI	**Code Repository:** GitHub tanujthakur9027-stack

Executive Summary: Commercial navigation systems (Google/Apple Maps) are fatally blind to flood water depth, debris blockages, and hospital bed saturation during natural disasters. In monsoon surges, ambulances unknowingly drive into 60cm submerged streets, causing engine hydrolock and critical patient entrapment. **RakshaPath** is an operational, full-stack disaster mobility intelligence network. It models physical vehicle wading limits, solves dynamic **Risk-Weighted A* safe corridors** around floodwater/rubble, and executes **Hungarian multi-criteria joint dispatch** reserving verified trauma beds before vehicle wheels turn. This document delivers complete technical evidence fulfilling all Stage-2 submission requirements.

1. Master 5-Minute Video Pitch Script & Storyboard (Rubric Item 1)

Use this second-by-second presentation script while recording the mandatory 5-minute video pitch. It aligns on-screen prototype interactions with spoken technical dialogue to score maximum marks across all evaluation criteria.

Timestamp	Screen Visual / Prototype Action	Presenter Spoken Dialogue & Key Talking Points
0:00 - 0:45 (45 sec)	**Camera on Presenter + Split Screen:** Show news clip of ambulance stalled in water, then cut to RakshaPath landing screen at http://localhost:5173.	**"Respected judges, during floods and natural disasters, standard GPS apps like Google Maps turn into fatal traps. Google Maps minimizes static road distance; it cannot 'see' whether a street is covered by 5 centimeters of rain or 70 centimeters of deadly floodwater. Ambulances drive blindly into submerged underpasses, destroying engines and losing critical patient lives. At Team Techies, we engineered RakshaPath (■■■■■■■): the autonomous emergency mobility network that calculates risk-weighted safe corridors and verifies hospital trauma beds in real time."**
0:45 - 1:45 (60 sec)	**Screen Recording: Top Bar & City Switcher** Click through City Jumps (Delhi NCR, Prayagraj, Lucknow, Dehradun). Toggle between **Tactical Dispatcher** and **Civilian Safety View**.	**"RakshaPath is not a mockup; it is a live, production-grade GIS platform. Notice our multi-state network covering major vulnerable basins across India. We built a dual-mode interface: For commanders, a tactical command dock; for citizens, an ultra-clean Civilian Safety Mode with zero confusing math, plain-language hazard bypass explanations, and 1-touch NDMA hotlines (112, 108, 1078). Observe the live telemetry ticker streaming real-time river discharge from CWC gauge telemetry stations."**

1:45 - 2:45 (60 sec)	**Screen Recording: Live Demonstration (Scenario 1)** Click ■ **Hackathon Tour** ➔ Select **Scenario 1: Gomti River Breach (Lucknow)**. Show the map render both the **Emerald Green Safe Corridor** and **Red Dashed Naive Path**.	**"Let us test a live scenario. The Gomti River has overflowed, putting the barrage link under 65 centimeters of water. An ALS ambulance has a maximum clearance of 30 centimeters. Notice what happens: The naive shortest route plotted in red dashed lines cuts straight through the 65cm hazard, which would submerge an ambulance. RakshaPath's physics engine applies an exponential penalty to the flooded segment and routes our ambulance via the elevated Shaheed Path flyover—achieving an 88% net risk reduction and guaranteeing 100% passability."**
2:45 - 3:30 (45 sec)	**Screen Recording: Vehicle Profile & Sliders** Switch vehicle from Ambulance to **NDRF Heavy Truck (85cm)** and **Flood Boat**. Move water depth slider from 30cm to 90cm. Click ■ **Launch Mission Dispatch**.	**"Our engine dynamically adapts to vehicle physics. When we switch mobility asset to an NDRF All-Terrain Truck (85cm wading limit) or tactical flood boat, the algorithm instantly re-evaluates segment traversability. Watch as we launch Live Mission Dispatch: An animated GPS beacon tracks along the corridor in real time with dynamic speedometers, countdown clocks, and live audio-tactile radar telemetry."**
3:30 - 4:15 (45 sec)	**Screen Recording: Hungarian Hospital Bed Match + Mission Brief** Show the destination hospital card with **12 Trauma Beds Verified**. Click **Mission Brief** to display the official NDMA dispatch manifest.	**"Routing is only half the battle. If an ambulance arrives at a hospital whose emergency ward is saturated, the patient dies. RakshaPath implements a Hungarian Bipartite Matching algorithm that jointly solves vehicle dispatch and reserves verified hospital ICU beds simultaneously. With one click, dispatchers export this official NDMA Mission Manifest for field records."**
4:15 - 5:00 (45 sec)	**Camera on Presenter + Slide Summary** Show 25.9% response time reduction graph, BOM table, 75-day Gantt roadmap, and Team Techies closing slide.	**"Through rigorous SimPy discrete-event simulation over 200 flood emergency calls, RakshaPath achieved a 25.9% reduction in P90 tail response time and 100% elimination of hospital diversion mismatches. Our client-side embedded engine deploys to Vercel in 1.49 seconds with zero server dependencies, guaranteeing offline resilience during power blackouts. Team Techies has delivered a battle-ready solution for national resilience. Thank you!"**

2. Technical Breakdown, System Architecture & IoT Schematics (Point 1 & 2)

RakshaPath bridges macro hydrologic forecasting with micro-level vehicle navigation. The platform operates on a three-tier decoupled architecture: IoT Edge Perception, Algorithmic Core, and Edge Delivery.

Subsystem Layer	Technical Component	Operating Principle & Engineering Mechanism
1. Perception & Telemetry	IoT Ultrasonic River Gauges + CWC / IMD Radar APIs	Solar-powered ultrasonic sensors (JSN-SR04T) mounted on river bridges poll water elevation at 30s intervals. Data is transmitted via LoRaWAN / 4G telemetry to ingest river crest rates (dh/dt).
2. Hydro-Physical Hazard Modeling	Physics Runoff Model + Soft-Sigmoid Passability	Predicts road segment inundation depth ($d_{ ext{water}}$) and cross-references against vehicle class clearance. Evaluates debris and damage multipliers to compute edge resistance.
3. Optimization Engine	Risk-Weighted A* Corridor + Hungarian Matcher	Computes dual Pareto-optimal routes (Safe Corridor vs Naive). Simultaneously pairs incident triage priority (P_1-P_4) to capable vehicle class and verified hospital bed capacity ($C_{ ext{trauma}}, C_{ ext{ICU}}$).
4. Edge Presentation	React 19 + MapLibre GL + Embedded Dijkstra Fallback	Renders 60fps GIS overlays with zero server roundtrip. Features offline embedded graph engine in Web Worker so routing continues if cellular towers collapse during storms.

IoT Field Telemetry Node Architecture (Hardware Blueprint)

To ensure ground-truth water elevation during communication blackouts, RakshaPath specifies an autonomous solar-powered IoT river gauge node installed at river bridges and highway lowlands:

Module	Component Specification	Functional Role	Power / Protocol
Microcontroller	ESP32-S3 Dual-Core Xtensa 240MHz	Edge data logging, Kalman filtering of wave noise	3.3V, Active: 80mA, Sleep: 15µA
Depth Sensor	JSN-SR04T Waterproof Ultrasonic Transducer	Measures water surface distance (20cm to 600cm range, ±1mm)	5V DC, 30mA pulse
Telemetry Transceiver	SX1262 LoRaWAN 868/915MHz + SIM7600G 4G Fallback	Long-range telemetry (up to 12 km non-line-of-sight to gateway)	+22dBm output, LoRaWAN Class A
Power Management	10W Monocrystalline Panel + 18650 LiFePO4 (6000mAh)	Continuous autonomous power for 14 continuous days without sunlight	MPPT solar charger (TP5100)

3. Detailed Mathematical Formulations & Simulation Validation (Point 3)

RakshaPath replaces heuristics with deterministic hydro-physical penalty equations and discrete-event queuing theory. Below are the governing equations implemented in the algorithm:

Equation 1: Vehicle Water Passability Multiplier (M_{water})

For standard terrestrial emergency vehicles with maximum wading clearance $d_{\max}$ (e.g., Ambulance $d_{\max} = 30\text{cm}$, NDRF Truck $d_{\max} = 85\text{cm}$):

- If $d_{\text{water}} \le 5\text{cm}$: $M_{\text{water}} = 1.0$ (Zero penalty, dry surface)
- If $5\text{cm} < d_{\text{water}} \le 0.5 \cdot d_{\max}$: $M_{\text{water}} = 1.0 + 0.8 \left(\frac{d_{\text{water}}}{d_{\max}}\right)$ (Mild hydrodynamic drag)
- If $0.5 \cdot d_{\max} < d_{\text{water}} \le d_{\max}$: $M_{\text{water}} = 1.5 + 3.5 \left(\frac{d_{\text{water}} - 0.5 d_{\max}}{0.5 d_{\max}}\right)$ (Submerged exhaust, high crawl resistance)
- If $d_{\text{water}} > d_{\max}$: $M_{\text{water}} = 5000.0$ (Critical hydrolock hazard; edge marked **IMPASSABLE**)

Equation 2: Risk-Weighted A* Edge Evaluation Function

The A* heuristic cost $f(n) = g(n) + h(n)$ evaluates composite traversal cost for every road segment $e = (u, v)$ with physical road distance d_e:

$$W_e = d_e \times \Big[M_{\text{water}}(e) \times M_{\text{damage}}(e) \times M_{\text{debris}}(e) \Big]$$

Where $M_{\text{damage}} = 1.0 + 1.5 (\text{damage}/\text{tol})$ and $M_{\text{debris}} = 1.0 + 1.2 (\text{debris}/\text{tol})$. If any threshold is breached, W_e spikes by $\times 5000$, forcing Dijkstra to dynamically detour around the obstacle.

Equation 3: Multi-Criteria Hungarian Joint Dispatch Cost Matrix

To jointly assign rescue units $u_i \in U$ to emergency triage calls $j \in J$ and route to verified hospital $h_k \in H$:

$$C(i, j, k) = w_1 \cdot T_{\text{transit}}(u_i \to j \to h_k) + w_2 \cdot \big(1.0 - R_{\text{corridor}}(e)\big) + w_3 \cdot \left(\frac{1}{B_{\text{trauma}}(h_k) + 1}\right)$$

Where weights are calibrated to $w_1 = 0.50$ (time), $w_2 = 0.35$ (risk avoidance), and $w_3 = 0.15$ (hospital bed saturation). Bipartite matching minimizes total global regret in $O(N^3)$ polynomial time.

Simulation Validation Logs (SimPy Monte Carlo Benchmark)

A 200-incident stochastic discrete-event simulation was executed under severe monsoon surge ($T+2\text{h}$). Performance of RakshaPath was benchmarked against the status-quo standard dispatch:

Metric Evaluated	Standard Dispatch (Google Maps + Nearest 108)	RakshaPath Optimization	Engineering Improvement
P90 Response Time (Tail)	38.2 Minutes	**28.3 Minutes**	**25.9% Faster Arrival**
Ambulance Submersion Rate	41.0% (Vehicles Stalled in >30cm water)	**0.0% (Zero Submersions)**	**100% Elimination of Stalls**
Hospital Diversion Rate	28.5% (Arrived at full ER)	**0.0% (Bed Pre-Reserved)**	**100% Elimination of Diverts**
Average Route Risk Factor	7.40x Hazard Multiplier	**1.08x Low Risk**	**88% Net Risk Reduction**
Vercel Edge Cold Start	N/A (Traditional Server: ~3.2s)	**0ms (Embedded Engine)**	**Zero-Latency Edge Execution**

4. Itemized Bill of Materials (BOM) & Unit Cost Breakdown (Point 4)

Commercial disaster platforms cost millions in proprietary hardware. RakshaPath uses commercial-off-the-shelf (COTS) components and serverless cloud architecture, drastically lowering unit deployment cost.

Category	Component Description & Model	Qty	Unit Cost (INR)	Unit Cost (USD)	Sourcing Viability
IoT Sensor Node	ESP32-S3 Microcontroller Unit + Custom PCB	1	■ 650	$ 7.80	Mouser / Robu.in (COTS)
IoT Sensor Node	JSN-SR04T Waterproof Ultrasonic Transducer	1	■ 480	$ 5.75	Amazon India / Robu.in
IoT Sensor Node	SX1262 LoRaWAN Transceiver Module (868MHz)	1	■ 750	$ 9.00	Waveshare / Semtech
IoT Sensor Node	10W Monocrystalline Solar Panel + TP5100 MPPT	1	■ 850	$ 10.20	Local Solar Vendors
IoT Sensor Node	6000mAh LiFePO4 Battery Pack + IP67 Enclosure	1	■ 950	$ 11.40	Local Enclosure Fabricators
Subtotal (Hardware)	**Complete Autonomous IoT River Gauge Station**	**1 Unit**	**■ 3,680**	**$ 44.15**	**100% Readily Available**
Fleet In-Cab Terminal	Ruggedized 8-inch Android Display Tablet (IP68)	1 / vehicle	■ 8,500	$ 102.00	GeM Portal / Samsung Active
Central LoRa Gateway	MikroTik wAP LR8 Outdoor Gateway (15km radius)	1 / district	■ 14,200	$ 170.00	MikroTik India Distributor
Cloud / Edge Hosting	Vercel Edge Global CDN + Serverless Storage	Annual	■ 0 (Free Tier / ■1,600 Pro)	$ 0 - $ 20/mo	Instant Cloud Setup

Cost Analysis & Scalability: A complete city pilot deployment (e.g., Lucknow with 12 critical bridges and 20 emergency vehicles) requires an initial hardware CAPEX of approximately ■ **2,14,160 ($ 2,570 USD)**. This is less than 2% of the cost of one emergency ambulance, making RakshaPath immediately viable for SDRF and municipal disaster budgets.

5. Team Roles & Task Execution Responsibilities (Point 5)

Team Name: Techies • Track: Diploma & Higher Education

Team Member & Title	Domain & Core Technical Skills	Assigned Project Responsibilities & Execution Scope
Tanuj Kumar Singh **Lead AI & GIS Systems Architect**	Python, NetworkX, LightGBM, Spatial Graph Theory, SimPy	Architected Risk-Weighted A* corridor algorithm, vehicle hydro-physical penalty equations, and discrete-event Monte Carlo simulation validation.
Co-Developer 1 **Full-Stack Edge & GIS Engineer**	React 19, MapLibre GL, Tailwind CSS, Vite, Vercel Edge	Engineered GIS map overlays, satellite layer switcher, collapsible HUD slider, offline embedded Web Worker router, and real-time mission telemetry tracking.
Co-Developer 2 **Embedded IoT & Sensor Systems Lead**	ESP32 C++, JSN-SR04T, LoRaWAN SX1262, Solar MPPT	Designed waterproof IoT river gauge station, sensor calibration routines, battery power budget, and LoRa-to-cellular gateway relay protocol.
Co-Developer 3 **Emergency Operations & Testing Lead**	Disaster Management, Field Testing, UI/UX, NDMA Protocols	Calibrated Indian city test graphs (Delhi, UP, Uttarakhand), designed Civilian Safety View, conducted user usability testing, and authored NDMA dispatch brief format.

6. Structured 75-Day Execution Milestone Roadmap (Point 6)

To advance from working hackathon prototype to municipal disaster deployment, Team Techies has formulated a rigorous 75-day delivery sprint broken down into 5 progressive phases:

Phase & Timeline	Milestone Deliverables & Engineering Objectives	Key Validation Gate	Current Status
Phase 1: Days 1–15 Algorithmic Core & Prototype	• Implement Risk-Weighted A* graph solver • Build multi-disaster vehicle profile equations • Develop React 19 MapLibre GIS frontend • Deploy embedded standalone engine to Vercel	Passing 22/22 pytest tests; <1.5s Vercel build time	**COMPLETED (100%)**
Phase 2: Days 16–30 Hardware PCB & IoT Lab Testing	• Fabricate custom ESP32-S3 ultrasonic sensor PCB • Calibrate JSN-SR04T transducer in hydraulic flume • Implement LoRaWAN SX1262 telemetry packet protocol • Conduct 7-day battery depletion stress test	Sensor reading accuracy ±2mm; Zero packet loss under rain test	**IN PROGRESS (65%)**
Phase 3: Days 31–45 Field Pilot Deployment	• Install 4 pilot solar gauge stations on Gomti River bridges • Integrate live IMD doppler radar rainfall API • Deploy ruggedized Android tablets inside 6 test ambulances • Establish LoRa gateway link to Lucknow command center	Continuous 99.8% telemetry uptime across 14 days	**SCHEDULED (Day 31)**
Phase 4: Days 46–60 Drill Simulation & Red Team Stressing	• Conduct live evacuation drill with civil defense • Simulate complete cellular tower blackout (offline mode test) • Benchmark Hungarian hospital bed pre-reservation with ERs • Perform cybersecurity penetration audit on API gateway	End-to-end corridor failover <300ms during simulated road collapse	**PLANNED**
Phase 5: Days 61–75 NDMA Integration & Handover	• Finalize API connectors for 112 National ERSS protocol • Deliver municipal operations training & SOP manual • Package containerized on-premise Docker deployment • Final submission for national disaster tech deployment	Formal signoff by regional Disaster Management Authority	**PLANNED**

7. Identified Technical Bottlenecks, Gaps & Engineering Challenges (Point 7)

Transparent engineering requires identifying real-world operational constraints and technical limits under extreme disaster stress:

Technical Challenge / Gap	Root Technical Bottleneck	Engineered Architectural Solution in RakshaPath
1. GPS Multipath & Signal Loss	Dense torrential cloudbursts and urban high-rises cause GPS drift (up to 40m error) or temporary satellite signal loss.	**Dead Reckoning & Map Matching:** Integrated Kalman-filtered vehicle odometry and topological edge-snapping algorithms to snap coordinates to verified road vectors even during GPS dropouts.
2. Sensor Drift & Debris Siltation	Turbulent river rapids, floating debris, and heavy silt create false echo returns on ultrasonic transducers.	**Multi-Sample Median Windowing:** Microcontroller executes 50-sample sliding window median filter with outlier rejection, ignoring transient wave peaks and debris splashes.
3. Telecom Infrastructure Collapse	Category-4 cyclones or severe floods knock down cellular towers and fiber backhauls, cutting internet connectivity.	**LoRa Mesh + Client-Side Embedded Graph:** High-priority sensor telemetry falls back to 868MHz LoRa mesh; tablets run fully offline A* routing without server roundtrips.

4. Asynchronous Hospital Bed Latency	Hospital staff during mass-casualty events may delay updating bed counts in centralized computer software.	**Dynamic Decay & Safety Buffer:** RakshaPath implements exponential reservation decay ($B_{\text{effective}} = B_{\text{reported}} - \lambda \cdot \Delta t$) to prevent phantom bed allocations.

8. Systematic Risk Matrix & Mitigation Protocol (Point 8)

To ensure human life safety and operational compliance during disaster deployment, a formal 4-dimensional risk mitigation matrix has been formulated:

Risk Description & Domain	Severity	Likelihood	Failure Mode Impact	Proactive Engineering Mitigation Protocol
R1: Inaccurate Flood Level (Hardware / Sensor)	**HIGH**	LOW	Ambulance dispatched into unexpectedly deep water.	**Triple Redundancy:** Satellite precipitation radar feeds cross-validate ground ultrasonic sensors. If gauge delta >25%, system defaults to conservative safety factor (2x penalty).
R2: Driver Ignoring Route (Human Factors)	**MEDIUM**	MEDIUM	Driver takes familiar flooded street instead of detour.	**Audio-Tactile En-Route Alerts:** Ruggedized in-cab terminal emits high-decibel audible alarm when vehicle deviates toward a submerged road segment.
R3: Complete Grid Blackout (Environmental)	**CRITICAL**	MEDIUM	Cloud server and power lines down across district.	**Solar + Vercel Edge Offline Mode:** Field IoT gauges operate on 14-day solar LiFePO4 batteries; mobile tablets run standalone embedded A* engine from cached PWA memory.
R4: Cyber Attack / False Alarm (Security / Network)	**HIGH**	LOW	Malicious false road blockages injected into network.	**Cryptographic Badge Verification:** Reports require HMAC SHA-256 digital signature from authenticated NDRF badge credentials; unverified reports require dual confirmation.
R5: Road Subsidence / Washout (Physical Infrastructure)	**CRITICAL**	LOW	Bridge structurally fails during flood crest.	**1-Tap Closed-Loop Crew Failover:** Dispatched field crews tap 'Report Road Blocked' in 1 second, immediately triggering automated reroute for all trailing emergency fleet.

Certification & Submission Signoff

Submitted By: Team Techies
Track: Diploma & Higher Education
Project: RakshaPath Autonomous Disaster Corridor AI
Verification Check: All 8 Rubric Points Fully Addressed
Prototype Readiness: Live on Vercel & Localhost:5173
Video Pitch: 5-Minute Script Ready for Recording